

DEREK WANG

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Education

University of Waterloo

Waterloo, ON, Canada

Bachelor of Applied Science in Computer Engineering

Expected April 2030

- Relevant Coursework: Fundamentals of Programming, Linear Algebra for Engineering, Project Studio

Work Experience

Youth Badminton Instructor

March 2024 – August 2025

City of Richmond Hill

Rouge Woods Community Center, Richvale Community Center — Richmond Hill, Ontario

- Coached a class of **14 beginner to intermediate students** (ages **9–16**) in **8-week terms**, focusing on footwork, technique, and strategy
- Designed a **new badminton curriculum** centered around **real game situations**, which was adopted by **all** badminton classes at Rouge Woods and Richvale, enhancing student engagement
- Provided personal **feedback and mentorship** while delivering lessons to improve **individual performance**, fostering an inclusive space

Projects

Waterloo Aerial Robotics Group | *Github*

August 2025 – Present

- Autonomy Team Member

Fruit Ninja AI | Python | YOLOv11 | *Source Code*

August 2025 – September 2025

- Created an automated pipeline to detect and cut fruits in the game Fruit Ninja, using **YOLOv11** object detection
- Organized a dataset of **816 images (514 training, 302 testing)** for the fruit detection model

TTC Escape | C# | Unity | *Source Code*

July 2025 – August 2025

- Competed in the 2025 **GMTK Game Jam**, created a game with the theme of "Loops"
- Utilized **Unity** to create a puzzlegame inspired by the Toronto Transit Commission with a **looping** mechanic

WebTyper | JavaScript | HTML | *Source Code*

March 2025

- Utilized **Javascript, CSS, and HTML** to turn any website into an interactive typing game
- Tracked past statistics such as **Words Per Minute(WPM), Accuracy, and total times played**

Checkmate Robotics Club | C++ | *Github*

September 2024 – May 2025

- Software team member of Team **16868C Rushdown** for the **VEX Robotics Competition**, invited to the **2025 VRC World Championships**
- Utilized **Okapi PROS Lib** to program robot functionalities, **Odometry** to track robot positioning, **PID Controllers** for precise motion control and error handling
- Wrote and tuned 15-second autonomous routes for matches, and 60-second programming skills, ranking **5th out of 80 teams** at **Provincial Championships**

Dr. X Robotics Academy | Java

July 2019 – May 2022

- Hardware team member of Team **16417 Dr. X Academy** for the **FIRST Tech Challenge**, invited to the **2022 FTC World Championships**
- Created **8 iterations** of the team's robot, including versions of the **drivetrain, arm, claw, turret** and a **backup robot**
- Wrote and tuned 30-second autonomous period for matches, using **java**, implementing **Dead Reckoning** for robot localization, and **PID Controllers** for motion control

Skills

Languages: Java, C++, Python, HTML5, CSS, JavaScript, C#, TypeScript

Developer/Engineering Tools: Git, Unity, Fusion 360

Libraries/Frameworks: ReactJS, OpenCV, YOLOv11, Tailwind CSS